

DARDI PEKA

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PROFESSIONAL SUMMARY

I am a graduate of Queen Mary University of London (First Class) and a postgraduate student at Imperial College London with first-hand experience in CFD-driven design, external aerodynamics, and simulation verification. Most of this experience is tied to my time at Queen Mary Formula Student. My individual research includes CFD simulations, such as a 2D URANS overset-grid study of tiltrotor wake aerodynamics with rigorous convergence checks. I am keen to apply my skillset in various engineering roles.

EDUCATION

Imperial College London - London, UK <i>MSc Advanced Aeronautical Engineering</i>	Sept 2026 – Present
Queen Mary University of London - London, UK <i>BEng Aerospace Engineering - First Class - 76.0% Overall (Y1: 76.8%; Y2: 75.8%; Y3: 75.9%)</i>	Sept 2023 – Jul 2026
Westbourne School - Penarth, Wales, UK <i>International Baccalaureate Diploma - HL: Mathematics AA, Physics, Computer Science</i>	Sept 2021 – Jul 2023

EXPERIENCE

Queen Mary Formula Student - London / Towcester, UK	Sept 2023 – Aug 2026
Head of Engineering	Jul 2025 – Aug 2026
- Led the engineering team of 7 department heads, 7 deputies and 40+ members to the team's first fully assembled car in over 2 years, ran weekly integration meetings and created the engineering Gantt chart to ensure departments stuck to the car-build timeline. - Maintained the master CAD assembly of the Formula Class car and resolved interface conflicts between chassis, suspension, powertrain, drivetrain, and electronics subsystems. Reviewed all engineering drawings for the competition submissions. - Oversaw and engaged in the redesign and rebuild of the wiring loom, running the CBR600RR engine that hadn't run in a year. Contributed to rear suspension hardpoint redesign for safer mounting and engine bay repackaging. - Wrote the team's Engineering Design Report (EDR) that covers the full vehicle concept and design decisions, scoring higher than the previous Formula Class. Prepared the largest engineering documentation handover to date.	
Head of Aerodynamics	Jul 2024 – Jul 2025
- Used STAR-CCM+ for simulation-driven design of the IC25 front wing, improving downforce by 2.2× while cutting component cost by 30% and saving 4.5 kg (from 11.5 kg) through modifying parametric CAD in SolidWorks. - Led the team through the FSUK 2025 design and cost events (covering for the Head of Engineering position): 9th in design, 5th in cost and 7th overall in the concept class, among 40+ competitors. - Led a 6-person department, providing CFD training for new members and simulation support for other departments. Managed the bill of materials and competition documentation, while ensuring there was detailed handover for the department deputy.	
Senior Team Member - IC Aerodynamics & Lap Time Simulation	Nov 2023 – Jul 2024
- Designed parametric front- and rear-wing CAD in SolidWorks, iterating to target aerodynamic performance. Co-presented suspension and chassis for the concept class (ranked top 5) with the Head of Suspension. - Developed the EV lap time simulation model using IPG CarMaker and OptimumLap (submission ranked top 7).	
Associate Team Member	Sept 2023 – Nov 2023
Completed a 4-week CAD (SolidWorks)/FEA (Ansys Workbench)/CFD (Ansys Fluent) training and analysis of a previous team's aero package to join the Senior Team.	

InvestIN Young Engineer Experience / 2Excel Engineering - London / Lasham, UK

Aug 2022

- Completed InvestIN's 5-day Young Engineer Experience (bridge-building and airfoil design challenges) alongside industry professionals from Rolls-Royce and BAE Systems.
- Assisted with glider assembly and observed launch operations. Gained insight into aircraft maintenance, safety and testing at 2Excel Engineering's MRO facility.

ENGINEERING PROJECTS

Integrated Tiltrotor Design Project - Queen Mary University of London

Sept 2025 – May 2026

Team project developing a modernised Urban Air Mobility tiltrotor based on the Bell XV-15; individual workstream on nacelle conversion mechanics and wake aerodynamics.

- Built a 2D URANS method in STAR-CCM+ for nacelle wake through a 45° conversion sweep (overset grid, chosen over mesh morphing by decision matrix), verified with a GCI study and a five-mesh sensitivity study (40k–700k cells).
- Found strong drag growth and lift sensitivity beyond ~30° tilt and low-band vortex shedding ($St \approx 0.13$ – 0.18), consistent with bluff-body wake literature (Q-criterion, λ_2 , FFT).
- Verified numerical credibility through a grid convergence (GCI) study in a simplified simulation case supported by first-cell-height and wall y^+ estimation.

Supersonic Wind Tunnel Experiment - Queen Mary University of London

Mar 2026 – Apr 2026

Individual coursework: schlieren analysis of a symmetric double-wedge airfoil at $M \approx 1.4$ in the QMUL supersonic wind tunnel.

- Used MATLAB with PIVlab for schlieren frame extraction and image processing; measured shock, expansion and Mach wave angles at 0° and 7.5° incidence with full uncertainty propagation (95% confidence intervals).
- Applied shock-expansion theory, oblique-shock relations, and Prandtl-Meyer functions to calculate surface pressures and force coefficients; identified the breakdown of attached-shock theory at 7.5° where the lower-surface shock detached.

Steering Wheel Design and Manufacturing - Queen Mary Formula Student

Dec 2025 – Apr 2026

- Designed the steering wheel in 2024 under the original driver set. Redesigned in 2025 for the new driver set while ensuring rule-compliance.
- Analysed the load path in Ansys Workbench and confirmed material choice using a decision matrix, picking CFRP over Aluminium. 3D printed the base and prototyped the wheel grips with the drivers.
- Early build using wet layup, rebuilt in 2026 using resin infusion in collaboration with the Deputy and Head of Composites.

IC26 Ride and Damping Study - Individual

Aug 2026 – Sep 2026

- Derived quarter-car equations of motion by hand, then built linear (straight road) MATLAB and nonlinear (asymmetric damping, bump stops, tyre-lift off due to kerbs) Simulink models to test ride and grip metrics for damping on metrics based on the QMFS IC26 Formula Class car.
- Demonstrated nonlinearity lowers the optimum damping ratio for kerb inputs. Reran the model to compare with available tyre stiffness data, correcting two secondary conclusions.

CERTIFICATES

- [CSWA - SolidWorks CAD Design Associate](#)
- [IBM Enterprise Design Thinking Practitioner](#)

TECHNICAL SKILLS

- **CFD & Simulation:** Simcenter STAR-CCM+, Ansys Fluent, RANS/URANS with overset grids, SST k- ω turbulence modelling, and verification/convergence studies
- **CAD & FEA:** SolidWorks parametric modelling, Ansys Workbench and SIMULIA Abaqus FEA
- **Vehicle Simulation and Dynamics:** IPG CarMaker, OptimumLap, MATLAB and Simulink
- **Post-Processing & Analysis:** MATLAB (PIVlab, data processing), FFT / frequency-domain analysis
- **Manufacturing:** Wet layups, resin infusion